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Implementation of lean techniques in the packaging machine to optimize the cycle time of the machine

Pravin Jadhav*, Nachiket Ekbote

Karmaveer Bhaurao Patil College of Engineering Satara, Maharashtra, 415001, India

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ABSTRACT

Lean manufacturing is one of the techniques that many major businesses are attempting to implement to stay competitive in the international market. This work aims to improve cost reduction, standardization, and assembly cycle time optimization of a packaging machine using lean techniques. Amongst various tools of lean manufacturing techniques like waste elimination, standardized work approach, visual management, etc., are used. Among the eight types of waste, waste related to the packaging machine like incorrect processing, waiting, and unused employee creativity were identified and reduced. Kaizen is done for value engineering purposes. The work instruction manual is created to avoid the waste of incorrect processing and optimize the machine's cycle time.

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1. Introduction

Lean manufacturing is a systematic approach to identifying and eliminating waste (non-value-added activities) through continuous improvement by flowing the product at the customer's pull in pursuit of perfection. Lean identified eight wastes in the manufacturing viz. overproduction, waiting, transportation, over-processing or incorrect processing, excess inventory, unnecessary movement, defects, and unused employee creativity. The system focused on identifying significant waste sources and using tools like JIT, production smoothing, and others to eliminate the waste. Lean manufacturing uses various tools like value stream mapping (VSM), 5S, visual management, kanban, Kaizen, line balancing, cellular manufacturing, just in time (JIT), single minute exchange of dies (SMED), standardized work, etc. to eliminate these wastes and achieve better productivity, smooth flow, reduction in cycle time and many benefits. The packaging is one of the critical sectors in any country. Vertical form fill seal machines play a significant role in food packaging. Every industry wants to improve its product and standardize it by removing all the non-value adding processes. Lean manufacturing techniques can be effectively used to remove the waste from the product's process and improvement [1].

Lean techniques can be useful in standardizing the product and achieve cost reduction. Nichrome India Ltd. has wide experience in providing innovative packaging solutions. They offer packaging solutions for specific product lines with high quality and flexible machine lines ranging from its food packaging machine lines to its liquid packaging machine product lines. The company, where lean manufacturing is dedicated to providing pouch client's requirements. The MAXIMA 400 machine is designed for a packaging **pouch** with a speed of 200 packs per minute. **The machine** has the flexibility to pack different products with different quantities on each head. This packaging machine is used for various products like milk, coffee, health drinks, instant mixes, premixes, coconut and hair dye, sugar, tea, seeds, snacks, biscuits, etc. **The machine** is a compact and versatile PLC-based control system. It has a standalone auto-tuning PID temperature **controller with a** pneumatically operated horizontal and vertical sealing system. Single or perforated pouches in a chain can be produced with this machine. The company needs to standardize this machine and eliminate the wastes related to this machine. Also, the cycle time is to be optimized.

2. Literature review

Extensive literature is available on lean manufacturing. Oliveira et al. [2] carried out work in a company dedicated to mechanical equipment production. The wastes were identified with VSM.

* Corresponding author.

E-mail address: pravin19580@rediffmail.com (P. Jadhav).

Nguyen and Hien [3] presented a process to transform a traditional production model to a lean one. A case study is carried out on lean techniques application in Re-engineering an electronics assembly line. Re-engineering emphasizes some aspects such as reducing wastes, standardizing work, internal logistics, workplace designing, and changing layout. Mlkva et al. [4] carried out work on standardization. By documenting the current best practice, standardized work forms the base for Kaizen or continuous improvement. Antonioli et al. [5] done a study in an automotive sector, more specially directed at production line manufacturing components for air-conditioning systems. The main objective was to standardize the operations, eliminate non-value adding processes, enhance productivity, and eliminate waste by continuous improvements. B. Kumar and M. Kumar [6] used lean manufacturing & line balancing tools to reduce the cycle time of an automobile plant, which contains many non-value added activities and work. It deals with studying the existing operation time for assembly line balancing to avoid the station delay. The implementation of lean tools is done, which resulted in a shortening of cycle time in an assembly line. Kamble and Kulkarni [7] calculated the cycle time and studied the existing method of different workstations, suggesting an improved approach to reduce the cycle time and improve productivity. The paper's advantages lie in enhancing productivity by reducing the cycle time, smooth flow of components for assembly, reducing the worker tiredness, and suggesting new methods and installing it. Malashree et al. [8] carried out to know the cycle time to produce a rolling shutter and propose a better method to reduce the cycle time using work-study techniques. They also recommended ways to decrease the worker's fatigue level by utilizing the ergonomics approach in the metal section industry. The industry's improvement was achieved by minimizing the cycle time, changing the operation sequence, and reducing the worker's fatigue. Kurdve et al. [9] worked on digitalized work instructions. In industry, standardization and quality assurance for manual work tasks are supported by digitalized work instructions. The study aims to select lean methods and equipment for digital assembly instructions and standardized work inspired by axiomatic design. Standardized work and green lean production systems are applied in a case study. Johansson et al. [10] presented a brief view of standardized work in Sweden's automotive industry. The paper emphasizes future research on implementing standardized work in international organizations focusing on local demands and cultural differences and similarities. Diaz et al. [11] focused on the cycle time study of aircraft assembly using a wing spar as a case-study. The lean philosophy was studied, and the current process was analyzed, and the results were obtained. The overall aim was to find potential areas for productivity improvement and suggest new solutions that would significantly decrease the cycle time. The study showed that it was possible and necessary to apply new techniques and technologies for manufacturing processes in the aeronautical industry. Mrugalska and Wyrwicka [12] presented the strict integration of humans in the manufacturing process, continuous improvement, and focus on value-adding activities by avoiding waste. Li et al. [13] tested experiments in which the assembly was supported by different work instructions with differing information content. The outcomes show that operator performance in perceived cognitive workload and information quality was affected by presented content in work instructions. Menna [14] gave a learning process planning approach that addresses special machinery assembly requirements, which had been designed. It was applied for the knowledge transfer regarding the assembly process of an integrally geared compressor. Learning process steps had to be adapted for each product, as every product in special machinery is unique. Mrugalska and Wyrwicka [15] presented the strict integration of humans in the manufacturing process, a continuous improvement, and a focus on value-adding activities

by avoiding waste. Industry 4.0 or the fourth industrial revolution emerged in the manufacturing sector, creating a smart network of products, components, machines, properties, individuals, and ICT systems in the entire value chain to have an intelligent factory.

Studies on implementing lean philosophy and improvement using various techniques like 5S, standardization, visual management, etc., are reported in many literature pieces. Moreover, work is observed in cycle time reduction by using line balancing techniques also. However, significantly less work was reported on implementing the lean methods for the company's particular product instead of executing the overall process. Less work is observed for the waste elimination that is the elimination of unused employee creativity by engaging the company's workers in the improvement of a product of the company. This present work aims to identify the wastes related to the packaging machine and find solutions to eliminate it and perform Kaizen and standardize the machine. Similarly, the aim is to improve the assembly process, optimize the machine's cycle time, standardize the machine assembly, and make the assembly flow smooth. This study is focused on standardization and cycle time optimization of a packaging machine.

3. Methodology

In this paper, a case study on a MAXIMA 400 packaging machine (shown in Fig. 1) is carried out. The methodology was carried in two phases. The first stage is to identify and eliminate the waste occurring in the machine's overall process and then analyze the

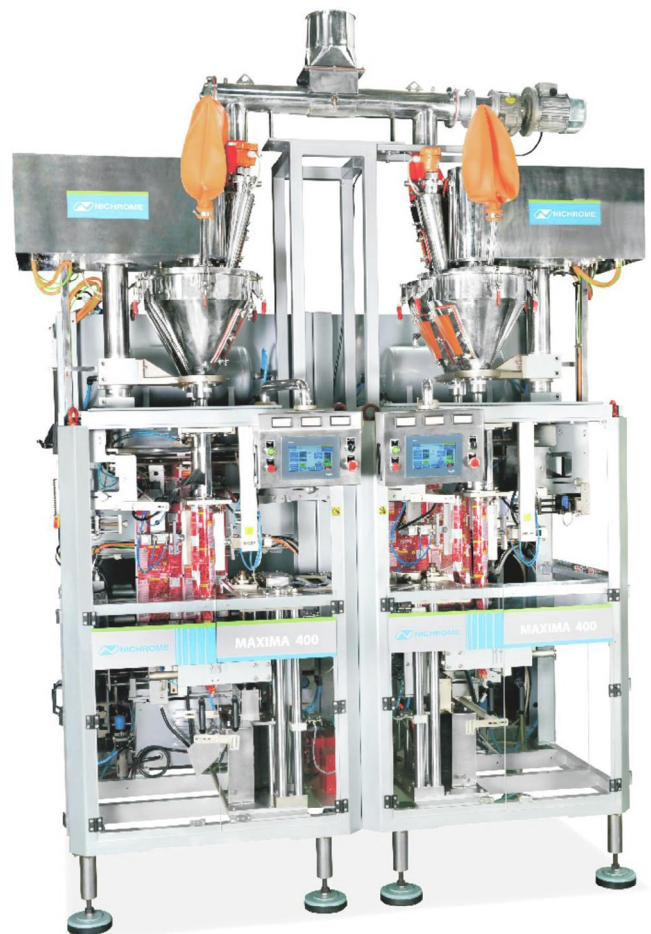


Fig. 1. MAXIMA 400 packing machine.

cycle time reduction through the application of Lean manufacturing through waste reduction, Kaizen & standardization, and cycle optimization.

3.1. Waste reduction

The methodology includes the machine study, and four wastes were considered from the eight forms of wastes. The unused employee creativity waste is a significant concern. As many times workers come up with creative ideas and improvements, they are not considered and implemented. This waste is taken into consideration, as shown in Table 1, and the machine study is done. The improvement points related to the performance, ease of assembly, trial, or cost reduction are noted by providing the workers with an engineering change suggestion form as per Appendix. According to their suggestion, the drawings of these parts are changed by the company's standard procedure. It is the step towards the standardization of the machine. Waste of unnecessary movement & waiting is also observed while performing the assembly operations on the shop floor. The workers, while doing the assembly, requires various tools and hardware. However, this less cost part plays a significant role in the assembly and requires much more quantity. The toolbox is always available to the workers, but the hardware is stored in a cupboard near the supervisor. And the workers have to go and take the proper hardware. The time is wasted in collecting the appropriate hardware as the workers don't read the assembly drawing and bring the hardware as per their experience. Also, there occurs unnecessary movement of workers if proper hardware is not found for searching it. That creates the unnecessary movement of workers and waiting, and hence much time is lost. Thus, a solution to eliminating this waste is providing a separate list of hardware for the particular assemblies that workers perform. Incorrect processing is the waste that occurs due to taking unnecessary steps to process the parts. It occurs due to inefficient processing due to poor tool and product design, causing unnecessary motion and producing defects. It is due to the lack of knowledge about the assembly. The assembly done is incorrect and may require disassembling and reassembling again, and thus the time is wasted. The solution to eliminating this waste was found out as providing the work instruction manual to the workers. Thus the assembly work instructions manual is created, which contains the main assembly is divided into various sub-assemblies and the exploded and assembled views from the software Solid Edge ST-8. So any worker or a layman can understand it and can perform the assembly efficiently.

Table 1
Design change suggestion by an employee.

Sr. No.	Nature of change	Reason for change
1	Valve, manifold mounting taps to be added, extra holes to be removed on the top plate	Holes removed due to product dripping from the top, value engineering
2	Panel bottom support to be modified, ribs to be repositioned	For panel mounting to be easy and wire routing purpose
3	M6 nuts are to be welded to the bottom side	Static box mounting to be easy
4	M6 taps for junction box, eight-pin cutout to be added, holes to be repositioned	Electrical wiring requirement
5	Tolerance to be changed	Easy fitment
6	Material, surface treatment to be changed, diameter 6mm diameter 8mm and 10 to 11	Resolve pin breakage issue
7	Hole dimension change from 6 to 8	Resolve pin breakage issue

3.2. Kaizen and standardization

Kaizen is performing continuous improvements, which is one of the essential parts of standardized work. Through the continuous improvement of your standard operations, your organization can systematically drive out waste & reduce costs. Thus Kaizen is done to reduce the manufacturing cost of the machine. Major assemblies of the machine were studied, and options were found out, and some unnecessary parts were removed to reduce the manufacturing cost. The Kaizen and the waste reduction caused the machine to standardize. Fig. 2 shows the machine structure before the change. It is the backside view of the machine structure. Fig. 3 shows the holes added for manifold mounting and the pneumatic valve mounting, which were not there on the drawing and used to be done on the workers' shop floor. Thus the waste of time on the shop floor required for drilling and tapping is reduced and eliminated. The cost increased in drilling and tapping work by the supplier is less and is compensated against the overall cost reduction in this structure assembly. Fig. 4 and Fig. 5 shows the structure for the MAXIMA machine in which the marked red cut-outs were present for mounting of the dowel pins and guide blocks assembly is removed as they are not at all required.

The pin is employed to fix the two machines, as shown in Fig. 5. At the customer end, the machines are not fixed to each other. Hence these are removed as we can only place the machines together with some gap in between. Also, the pins are drawn as if any of the machines have issues. Then it may transfer the vibrations to the other machine and create problems in pouch quality. And thus, the cost reduction is also achieved. Fig. 6 shows the drive table assembly with a horizontal sealing assembly. The assembly parts are bought out from suppliers, and the assembly is done on the company's shop floor. The parts top plate, vertical table plate,

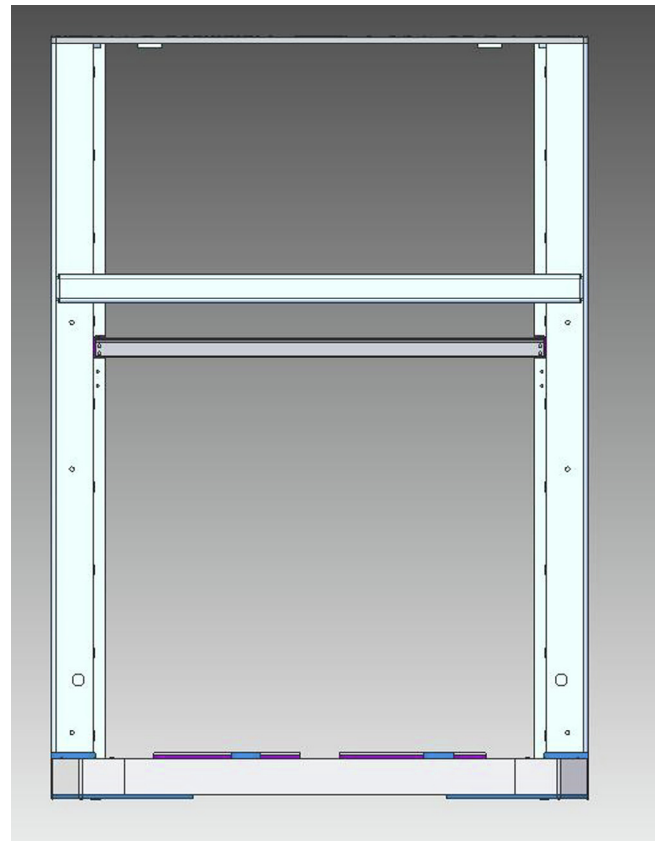


Fig. 2. Maxima structure before the addition of holes.

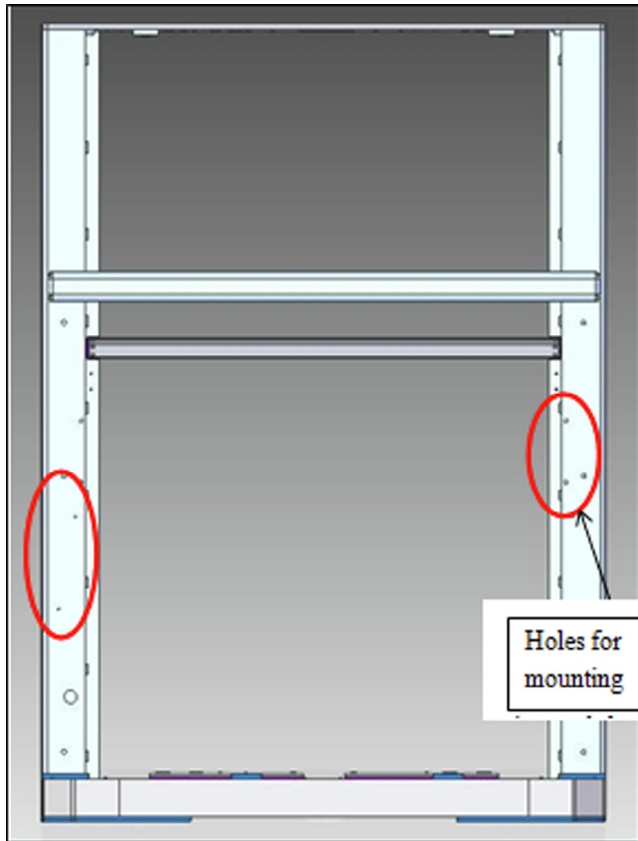


Fig. 3. Maxima structure after addition of holes.

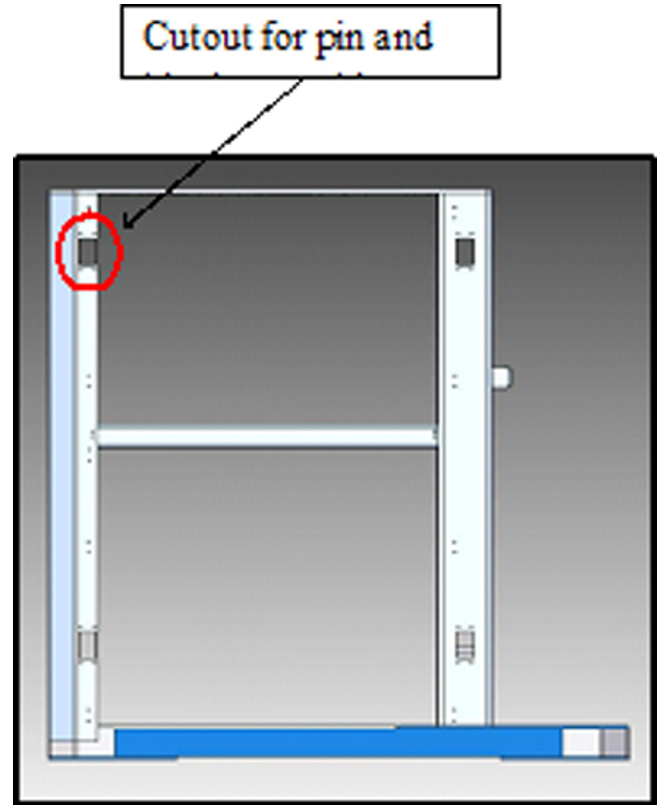


Fig. 5. Maxima structure left-hand side view.

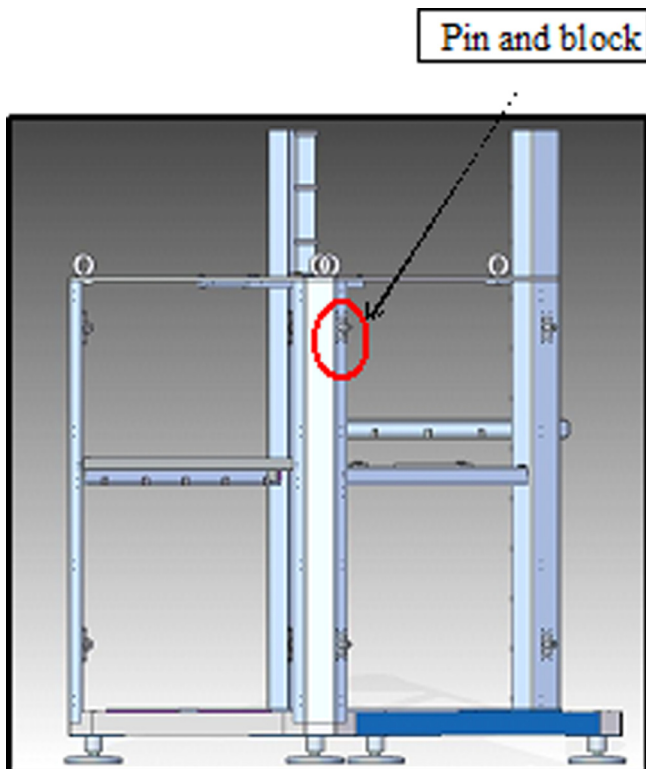


Fig. 4. Maxima structure before the change.



Fig. 6. Drive table assembly – before (ENP treated parts).

and the table mounting plate have the surface treatment as electroless nickel plating (ENP). But the other major visible parts of the machine are painted. So it was not looking aesthetically

pleasing. It was not at all required to have ENP treated components. Also, ENP treatment is very costly. Thus the surface treatment of these parts is changed from ENP to painting/powder coating as shown in Fig. 7. Thus the cost reduction/ value engineering is achieved for this assembly.

3.3. Cycle time optimization

Cycle time also can be optimized by adequately studying the assembly process of the machine. It is observed that many factors are contributing to waste generation and also increase the time of assembly. Hence, it is needed to focus on the causes that produce the actual cycle time to analyze and reduce, as shown in Fig. 8. Fig shows four levers: Manpower, components, methods, and tools with their consequences that contribute to actual cycle time. These levers are analyzed below:

- i. **Methods:** The analysis shows that the process is manual. There is a lack of standardization and visual management.
- ii. **Manpower:** Operators have freedom of movement in the workstation. Also, the Maxima machine is a costly machine with fewer orders than the company's other standard machines. Thus, workers get confused and cannot recollect the assembly process while performing the machine's assemblies as they don't read the critical assembly drawing.
- iii. **Components:** Operators sometimes have lead time because of the elements with low quality, lack of standardization, and the delay by the supplier. Rarely workers have to rework the component in the factory itself. The hardware required for assembly is kept in a rack present at the supervisor's table. Workers don't usually read the assembly drawing and use the hardware by their thumb rule. But this many times wastes their time to collect the proper hardware. Also, time is wasted in finding the appropriate hardware if not present in the rack. Thus the unnecessary movement of the worker happens looking for the hardware.



Fig. 7. Drive table assembly – after (Powder-coated parts).

- iv. **Tools:** Tools used are basic, low technology. As only assembly operation is done in the factory, essential tools like Allen keys set, spanners, etc., are used. Each worker has his toolbox containing these tools. In a scarce case, a worker has to wander in the factory looking for the proper tool.

Thus the above analysis for cycle time shows that the cycle time can be improved and optimized if we reduce the waste in the assembly shop. The actual assembly process is studied, and the approximate time is taken to observe and study the actual cycle time. The day to day work on the Maxima machine's shop floor is noted for an order of the machine. The factory's overall assembly process is not as that of the mass production industry's assembly process. Thus the stopwatch time cannot be taken, and hence general observation is done, and time analysis is done as per the sequence of machine assembly. Table 2 below shows the task performed, the duration required for the particular task.

After studying the machine's cycle time, it is observed that only two significant assemblies are performed on the company shop floor, and other assemblies are bought out from the suppliers. These two assemblies are time-consuming. These assemblies are required to be performed keenly and accurately as the assemblies are the machine's hearts. These assemblies may go wrong if not properly observed and executed and needs to be disassembled and reassembled again and thus increases the overall machine cycle time as shown in Fig. 9a and b. There is a requirement for proper guidelines and assembly procedures for performing these assembly operations. Hence, the assembly work instructions created to reduce the waste of incorrect processing can help optimize the machine's cycle time.

4. Results & discussion

After analyzing the packaging machine, it is observed that there is a lot of waste occurring in the machine's overall production process right from the design stage to the manufacturing stage. Waste reduction and standardization of machine waste identification are made to achieve this. Among the eight types of waste in lean manufacturing, wastes found were: unused employee creativity, unnecessary movement, waiting, and incorrect processing.

- i. **Unused employee creativity** waste is also of significant concern. This waste is taken into consideration in this study, and efforts were taken to reduce the same. Thus it achieved value engineering and the improvement in the machine and increased the easiness of doing the assemblies or reduced the onsite problems and increased manpower efficiency.
- ii. **Waiting and unnecessary movement** occurs for the searching of the hardware especially. That created waste and much of the time is wasted. This waste is reduced by providing workers the list of hardware required for the assemblies.
- iii. **Waste of incorrect processing** was due to incorrectly performing the assembly operations because of a lack of knowledge about the assemblies. It caused the assemblies to dismantle and reassemble again. That causes a waste of time. This waste is eliminated by providing work instructions to the workers. The assembly work instructions manual was created, which contains the main assembly is divided into various sub-assemblies, and the exploded and assembled views from the software Solid Edge ST-8. That helped any worker or laymen understand the assembly and can perform the assembly efficiently. This document enabled them to do the assembly correctly, with fewer defects, and in less time.

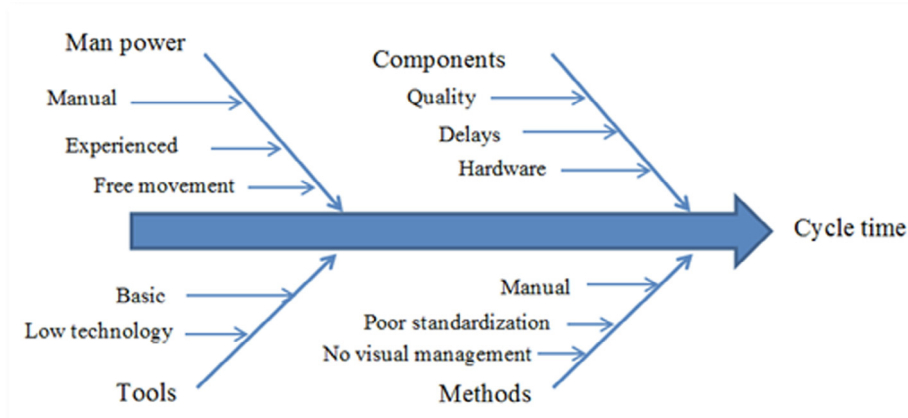


Fig. 8. Assembly process cause-effect diagram.

Table 2

Cycle time of maxima machine.

Task	Duration
Structure assembly received from a supplier	1 hr
Trolley assembly received from a supplier	1 hr
Drive table and horizontal sealing assembly	7 days
Leg fitment to structure	1 hr
Trolley assembly fitted to the structure	1 hr
Vertical seal assembly	5 hrs
Slitting assembly	3 days
Slitting assembly fitted to the structure	0.5 hrs
Unwinding/looping motor assembly fitted to structure	0.5 hrs
Panel box received and fitted	1 hr
Paper forwarding front received	0.5 hrs
Carrier arm fitted	0.5 hrs
Drive table mounting plate fixed, and wiring started	0.5 hrs
Paper forwarding front fitted to the structure	0.5 hrs
Vertical sealing assembly fitted to the structure	1 hr
Drive table assembly fitted into the structure	0.5 hrs
Duplex servo auger assembly received	1 day
Duplex servo auger assembly centering work	1 day
Duplex servo auger filler assembly fitted on structure	2 hrs
Former assembly fitted	1 day
Splicing table fitment	0.5 hrs
Wiring and pneumatic assembly work	2 days
Guarding assembly	1 day
Trial	6 days
Machine packed	1 day
Machine idle- ready to dispatch	5 days
Machine dispatch	1 day

- iv. The Kaizen performed resulted in the reduction of unnecessary parts and unnecessary processed parts. That caused the reduction of the manufacturing cost of the machine and the value engineering. The assembly work instruction manual provided also resulted in saving the assembly time, and thus the total cycle time was optimized.

5. Conclusions

The study is focused on the implementation of lean tools and techniques in the packaging industry. The paper presents improvement, cost reduction, standardization, and assembly cycle time optimization of a packaging machine using lean techniques. Amongst various tools of lean manufacturing techniques like waste elimination, standardize work approach, visual management, etc., are used.

In the research work, Kaizen is done by machine parts/assembly modification at the design level and implemented to reduce waste and cost-saving. Waste of unused employee creativity is elimi-

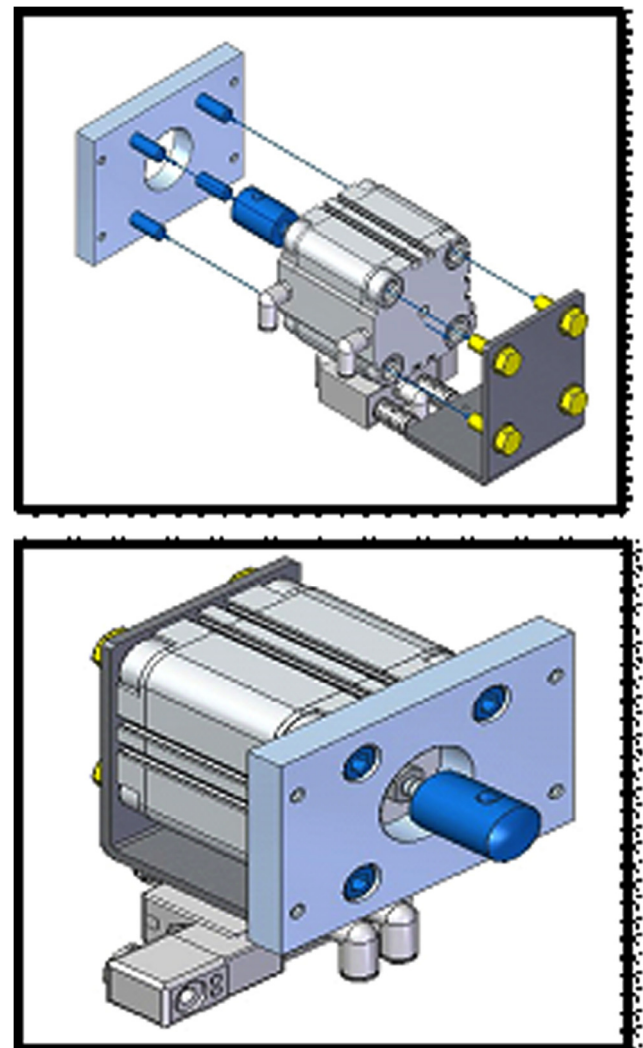


Fig. 9. a. Exploded view of a sub-assembly. b. Final views of a sub-assembly.

nated as workers were involved in suggesting the changes or modifications. Also, by studying the current assembly process and sequence, the assembly procedure document is prepared to increase workers' easiness for assembly and reduce the waste like incorrect processing and defects. The waste of waiting and unnecessary movement is overcome by providing required data like a list

of hardware to the workers. That reduced the time of completing the assemblies, and the cycle time is thus optimized. The tools like solid edge, MS Excel, etc., were used here in this work. From the research work, it can be concluded that lean manufacturing techniques can be used in any type of industry and any type of product to reduce the waste related to the manufacturing of that product. Unused employee creativity is a waste to be considered by every sector. Every industry should try to reduce the same by employee engagement programs, as it improves product improvement. Waste like incorrect processing can be reduced by visual management techniques, which also achieve cycle time optimization.

CRedit authorship contribution statement

Pravin Jadhav: Conceptualization, Investigation, Methodology, Supervision, Writing - original draft. **Nachiket Ekbote:** Data curation, Project administration, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Blank and filled engineering change suggestion form

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